

Project 3: Synthetic-to-Real Chessboard Image Generation from FEN

Final Report

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Repository: https://github.com/TalPodgor/DL_Project3

Abstract

This project addresses chessboard image generation from a FEN position and a requested camera viewpoint. The required output is a clean synthetic chessboard image and a more realistic image that preserves the same board state. We treat the problem as controlled synthetic-to-real image translation. Early unpaired translation with CUT/CycleGAN-style training produced realistic texture but did not reliably preserve exact board structure. The final system therefore uses a paired, pix2pixHD-style conditional generator, augmented with chess-specific semantic and geometric conditioning from Blender: an RGB render, a FEN semantic layout, a rendered silhouette, a depth-like map, and piece-edge cues. The selected checkpoint, chess_v5_bright_silABC, is trained on a camera-aligned Blender dataset with per-game oblique elevations of 38-44 degrees. On 140 held-out game-2 boards it achieves 0.922 square accuracy, 0.992 occupancy accuracy, 0.804 occupied-piece type accuracy, and 0.757 whole-board occupancy exactness under an independent square classifier. Qualitatively, the model gives the best practical trade-off between realistic wooden-board appearance and board-state preservation, although piece topology artifacts remain a central limitation.

1. Introduction

The task in Project 3 is to implement `generate_chessboard_image(fen, viewpoint)`. Given a FEN string and a viewpoint string, the function must save `synthetic.png`, `realistic.png`, and `side_by_side.png` under `./results/`. The generated realistic image should look like a photograph of a wooden chessboard while preserving the exact piece placement and the requested white-side or black-side view.

This task is harder than ordinary style transfer. A visually plausible chessboard is not sufficient: a pawn cannot appear on an empty square, a rook cannot become a bishop, and the camera orientation must remain consistent with the input viewpoint. At the same time, the real training photographs are low resolution, strongly rectified to the board plane, and contain small, low-contrast pieces. Tall pieces lean across square boundaries after rectification, so the board grid can be aligned while piece heads and bodies are not perfectly aligned.

Main contributions

- A FEN-controlled Blender rendering pipeline that creates synthetic chessboards and auxiliary control maps.
- A paired pix2pixHD-style translation model that conditions on RGB, semantic, depth, silhouette, and edge information.
- A camera-aligned V5 dataset that replaces near-top-down synthetic supervision with oblique Blender renders closer to the real photos.
- A set of quantitative and visual ablations showing why unpaired translation, silhouette-only conditioning, geometry locking, and naive parallax stretching were rejected.

2. Related Work

Image-to-image translation is commonly approached with conditional GANs. Pix2pix uses paired supervision and a conditional discriminator to learn a mapping from an input image to a target image. Pix2pixHD extends this idea with stronger high-resolution synthesis components, including multi-scale discrimination and feature matching. CycleGAN and CUT are unpaired approaches: they can learn domain style without one-to-one pairs, but their structure preservation is weaker for tasks that require exact object

layout.

Our final method is closest to pix2pixHD: it uses paired supervision and a conditional generator, but adds chess-specific inputs rather than relying on RGB alone. CUT and CycleGAN-style training were useful early baselines, but they were not selected because the task is not only to make synthetic chessboards look real; it is to keep the exact chess position.

3. Data and Preprocessing

The final training set is `datasets/chess_v5_oblique_aligned_bright`. It contains paired images where the left half is the Blender-rendered source and the right half is the real target. Each sample also has a semantic silhouette image and a depth-like image. The split contains 736 training pairs from games 4, 5, 6, and 7, and 140 held-out test pairs from game 2. The held-out game split is important because it tests whether the model generalizes to unseen positions and viewpoint instances rather than memorizing one board sequence.

A central empirical finding was that camera geometry matters. The older V5 oblique dataset used a global elevation of approximately 54.9 degrees, while the real board photographs were more oblique. The final aligned-bright dataset uses per-game elevations of 38, 40, 42, and 44 degrees with a mean of 40.7 degrees. This reduces the mismatch between the synthetic piece projection and the real rectified photographs.

Split	Games	Pairs	Notes
Train	4, 5, 6, 7	736	Paired Blender source, real target, semantic silhouette, depth
Test	2	140	Held-out game used for quantitative and qualitative evaluation

Dataset	Camera elevation	Resulting role
<code>chess_v5_oblique</code>	54.9 deg global	Rejected as too top-down for real piece geometry
<code>chess_v5_oblique_aligned_bright</code>	38-44 deg per game	Final training data for <code>bright_silABC</code>

4. Method

The final model is called `paired_geom_hd` in the codebase. It is a paired conditional image-to-image model in the pix2pixHD family, not a CycleGAN. The generator is a ResNet 9-block network with instance normalization. The discriminator is a multi-scale PatchGAN-style discriminator, and a local piece-crop discriminator is used during training to put additional pressure on occupied squares.

Input representation

For each position, the model receives a tensor formed by concatenating: (1) the RGB synthetic render, (2) a one-hot semantic map, and (3) three geometry channels. The semantic map uses `fen_silhouette` mode: full-cell FEN information protects board occupancy, while rendered silhouette pixels provide a more accurate outline of visible piece shapes. The geometry channels are a depth-like render, a piece silhouette mask, and a silhouette boundary channel.

Training losses

The generator is trained with adversarial loss, discriminator feature matching, VGG perceptual loss, masked L1 reconstruction, a silhouette-edge loss, contextual loss, a frozen square-classifier loss, and local crop-level piece losses. The masked L1 term uses a reduced weight on piece regions to avoid forcing pixel-perfect alignment where 3D parallax makes the real and synthetic piece heads differ. This is important because hard pixel losses on misaligned tall pieces tend to create blur or double-head artifacts.

Parameter	Final value
Model	<code>paired_geom_hd</code>
Dataset mode	<code>v5_oblique</code>
Semantic source	<code>fen_silhouette</code>
Generator	ResNet 9-blocks, ngf=64, instance norm
Batch size	1
Epochs	40 + 20 decay

Parameter	Final value
lambda_GAN, lambda_feat, lambda_VGG, lambda_L1	1.0, 10.0, 3.0, 7.5
l1_piece_w	0.7
lambda_edge, lambda_cx	1.0, 0.5
Local piece losses	piece GAN=1.0, feature matching=10.0, VGG=5.0

At inference time, the model still satisfies the Project 3 contract: only the FEN and viewpoint are needed. The system renders the synthetic board from the FEN, derives the same control maps from that synthetic state, forwards the model, and saves the required PNG outputs.

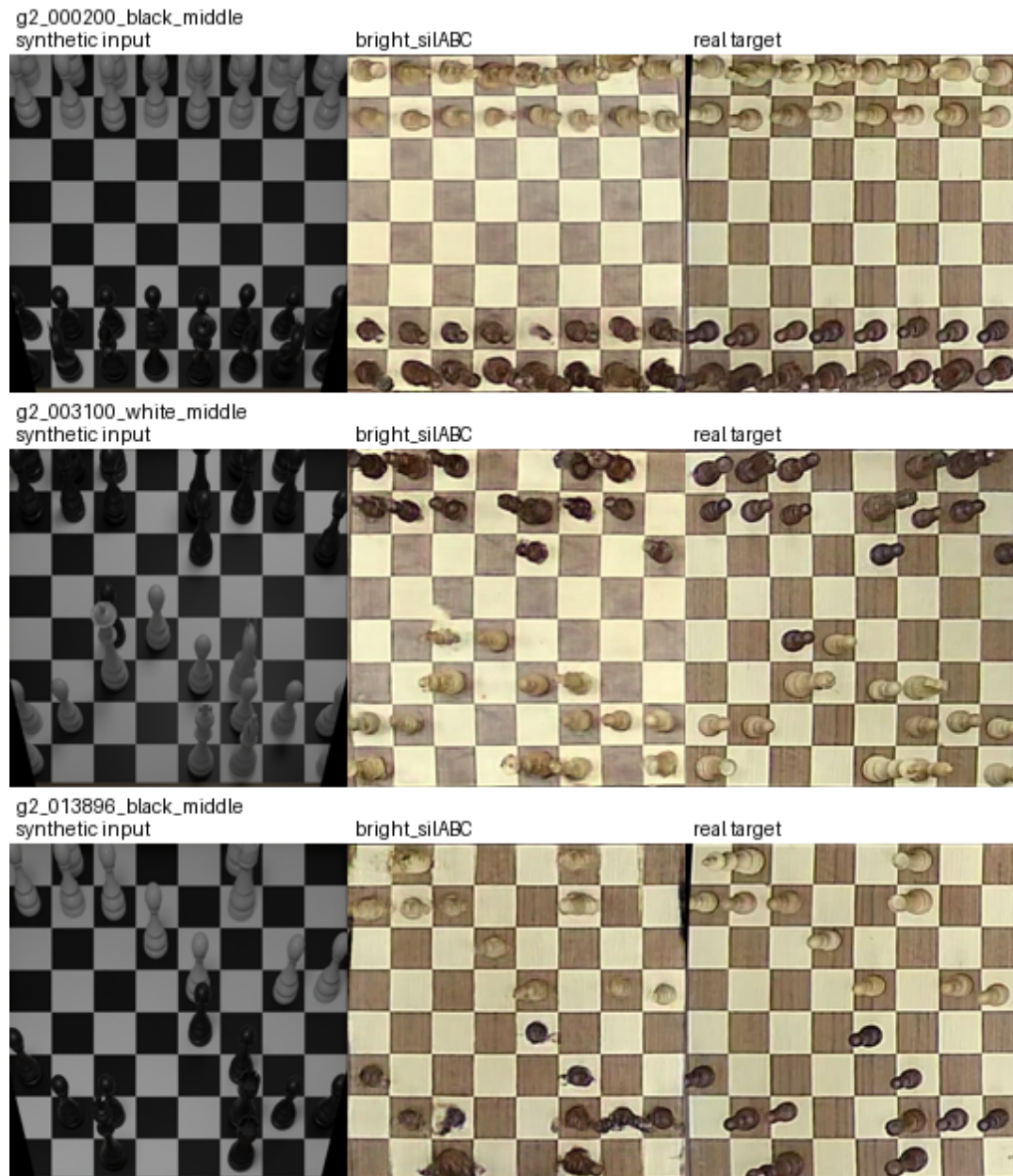


Figure 1. Final qualitative examples. Each row shows the Blender synthetic input, the selected bright_silABC output, and the real target image from the held-out test game.

5. Experiments

We evaluate using both automatic and visual criteria. The main quantitative evaluator is an independent square classifier trained on real board crops. It reports square accuracy, occupancy accuracy, occupied-piece type accuracy, phantom and missing-piece rates, and whole-board exactness. These metrics are useful for board-state preservation, but they are not sufficient for final visual quality because a merged blob can sometimes be classified correctly. Therefore we also use visual grids and specialized audits for double-head, halo, transparency, and piece-detail artifacts.

Run	Square	Occupancy	Type	Whole-board occupancy	Phantom	Missing
bright_silAB	0.9160	0.9924	0.7849	0.6429	0.0057	0.0109
bright_silABC (selected)	0.9221	0.9916	0.8040	0.7571	0.0057	0.0130
silAB noPieceD	0.9314	0.9942	0.8240	0.7643	0.0052	0.0068
pcomp_srcshape	0.9235	0.9955	0.7979	0.7929	0.0019	0.0090
geometry lock train	0.6993	0.8738	0.3075	0.0000	0.0293	0.2991
parallax8 fine-tune	0.9094	0.9846	0.7879	0.4500	0.0167	0.0130

The selected model is not the winner of every scalar metric. For example, noPieceD has slightly higher classifier scores. However, final selection used a stricter practical criterion: the model must preserve a realistic appearance and not introduce severe synthetic, milky, or transparent piece artifacts. Bright_silABC was selected as the best overall visual trade-off.

6. Ablation Study

The final method contains several components, so we performed ablations and negative tests to justify the design. The most important conclusion is that improving board-state metrics alone is not enough. Components that force exact geometry can remove double heads, but often destroy the photo-like style. Components that preserve style often leave the topology problem unsolved.

Ablation	Hypothesis	Observed result	Decision
Silhouette-only semantics	Visible silhouettes should encode piece layout	High phantom rate in the V5 probe; no whole-board occupancy exactness	Reject; full-cell FEN is necessary
Full-cell FEN + V5 geometry	FEN layout protects occupancy while geometry provides shape	Probe reached 1.0 occupancy on 16 boards	Use as V5 baseline
No piece classifier	Classifier loss might overfit or reward blobs	Improved some classifier metrics but did not clearly solve visual defects	Not final
Geometry lock	Hard source geometry will prevent double heads	Double-head defects dropped strongly, but pieces became washed and synthetic	Reject for final style
Two-stream / piece composite	Separate foreground and background generation	Style remained reasonable, topology mostly unchanged	Not enough
Parallax8 source stretch	Synthetic pieces are still too short or top-down	Worse occupancy, worse phantom rate, no double-head improvement	Reject

Run	Double-head score	Any-extra rate	Pawns any-extra	Officers any-extra
bright_silAB	2.4544	0.5342	0.5932	0.4481
bright_silABC	2.4004	0.5463	0.6236	0.4336
parallax8	2.5367	0.5593	0.6346	0.4496

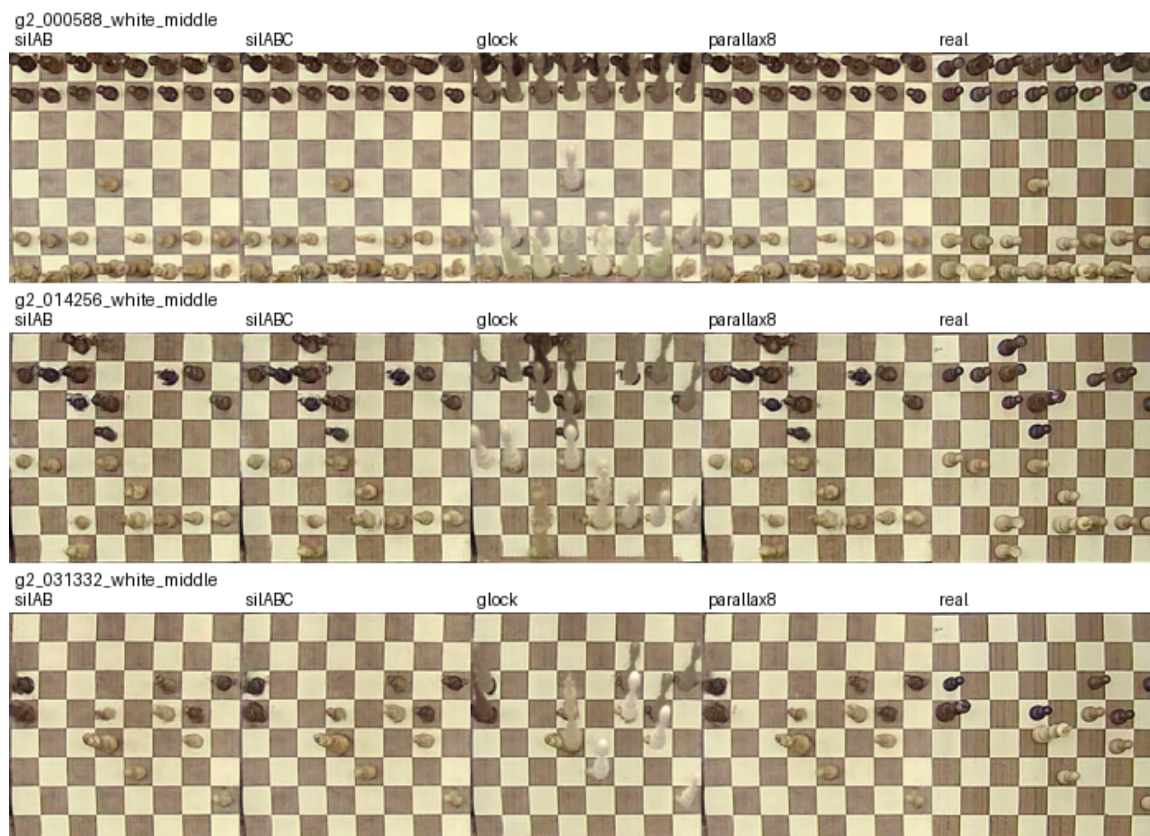


Figure 2. Qualitative ablation comparison. Geometry-lock improves structure but looks synthetic; parallax8 does not improve the selected bright_silABC model.

7. What Did Not Work

Unpaired translation was the first major failure mode. CUT/CycleGAN-style models can transfer wooden-board style, but they do not receive a strong pairwise penalty for moving pieces or inventing piece-like texture on empty squares. This is unacceptable for chess, where the state must remain exact.

A second failure mode was naive geometric correction. The final aligned-bright dataset already uses camera elevations close to the real photos. A later parallax8 experiment stretched the synthetic piece silhouettes upward before fine-tuning from bright_silABC. It reduced neither the double-head audit nor the visual artifacts, and it degraded occupancy exactness from 0.757 to 0.450. This suggests that the remaining mismatch is not solved by a simple post-render stretch.

A third failure mode was hard geometry locking. It made double heads much rarer, but it achieved this by injecting synthetic-looking piece structure into the output. The resulting pieces looked washed, transparent, or milky, which failed the realistic-image requirement.

Finally, automatic mask extraction with SAM and color trimming did not produce clean real-piece sprites. At the available 40-60 pixel piece scale, many white-piece pixels have almost the same color as light board squares. Automatic segmentation often captured the square or fragmented the piece, so it was not used in the final model.

8. Discussion and Limitations

The selected model is a practical submission model, not a complete solution to realistic chess piece synthesis. It generally preserves occupancy well and creates a visually plausible wooden board, but local piece geometry remains imperfect. Common failures include blob-like pawns, side-lobed heads, weak separation between neighboring pieces, and occasional phantom texture on empty squares.

The main limitation is the supervision itself. The real targets are board-rectified photographs of a 3D scene. A homography aligns the board plane, but not the elevated heads of pieces. Pixel and perceptual losses therefore supervise slightly incompatible shapes. Reducing L1 pressure on piece regions avoids severe blur, but also leaves the generator with weaker direct supervision for fine piece structure.

The most credible future improvement would be new supervision rather than more loss tuning: manual annotation of real piece masks, higher-resolution capture, or a renderer/camera calibration pipeline that

matches the real camera pose and lens more precisely. With such data, one could train a dedicated foreground matting or piece-refinement model instead of asking a small GAN to infer ambiguous local topology from low-resolution photos.

9. Conclusion

We built a controlled synthetic-to-real chessboard generation pipeline for Project 3. The final model is a paired pix2pixHD-style generator with FEN, silhouette, depth, and edge conditioning. The selected bright_silABC checkpoint is trained on the camera-aligned bright V5 dataset and gives the best overall trade-off between realistic appearance and state preservation among the tested models. The project also produced a clear negative result: under the current low-resolution data and automatic supervision, there is a persistent trade-off between clean piece topology and realistic photographic style. This limitation is documented through both quantitative ablations and visual comparisons.

References

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